

SURYA'S EXECUTIVE REFERENCE

FDE Capstone Bible

Cell and Gene Therapy
Patient-to-Batch Orchestration

One fixed story.

Understand the client ask, the 21-stage operating model, the working demonstration, the evidence and the honest lifecycle decision.

Prepared for Surya

Version 1.0 | 15 September 2026

FDE-Final-Capstone | Synthetic academic POC

[START HERE](#)

How to use this bible

Do not try to memorize every artifact. First identify the type of question, route it to the correct stage, give the short answer and then point to the evidence.

1. Hear the keyword

Examples: DDD, C4, risk, data, test, deployment or ROI.

2. Route to a stage

Use the five-phase memory map and quick question index.

3. Answer in four parts

Yes or no; stage; what we did; evidence and limitation.

THE ANSWER FORMULA: Yes or no -> Stage -> What we did -> Evidence -> Honest limitation

Use the sections in this order:

- Pages 3-6: understand the business story and solution flow.
- Pages 7-12: understand all 21 FDE stages in five phases.
- Pages 13-17: understand the POCs, architecture, AI boundaries and evidence.
- Pages 18-22: prepare the live demo and answer reviewer questions.
- Final pages: limitations, lifecycle decision, glossary and one-page memory sheet.

Golden rule

Never convert an academic POC result into a production, clinical, regulatory or financial claim. State what was demonstrated and what still requires external evidence.

EXECUTIVE SUMMARY

The capstone in one page

Build one safe, evidence-driven view of the journey from a specific patient to that patient's manufacturing batch.

What is CGT?

Cell and Gene Therapy (CGT) uses cells or genetic material to treat serious diseases. In this capstone's patient-specific journey, cells are collected for one patient, transported, manufactured into a treatment, Quality tested and returned for treatment.

Why is this difficult?

Clinical, scheduling, logistics, manufacturing, laboratory and Quality teams use different systems. Evidence can be delayed, missing or contradictory. A wrong identity link, duplicate reservation or unsupported release state can create serious risk.

What was delivered?

One Control Tower A controlled view of patient, batch, evidence, exceptions and decisions.	Three working POCs Identity readiness, safe slot orchestration and Quality release.	One honest decision Accept the academic POC; restrict real-data and production use.
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The result

All 21 FDE stages contain artifacts. All 77 automated tests pass. Fifty-five of 57 evaluation cases pass with zero failures; two human-study cases remain inconclusive. Twenty-five of 27 requirements are internally verified.

BUSINESS NEED

What the client is asking for

The client needs a reliable way to answer five questions:

#	Client question	Required proof
1	Is this the correct patient?	Identity and chain-of-identity evidence
2	Is the patient ready for manufacturing?	Consent, eligibility, site and authorization evidence
3	Was the manufacturing slot reserved?	Command state, external result and reconciliation
4	Can the batch be released?	Manufacturing, QC, deviation, thermal and Quality authority
5	Why should we trust the answer?	Citations, source provenance, decision owner and audit history

The engineering problem

The fragmented environment cannot always determine the current patient-to-batch state, why it is true, who is authorized to decide and what must happen next.

Identity risk Conflicting identifiers can create an unsafe patient link.	Operational risk Timeouts and retries can create duplicate external actions.	Authority risk Missing evidence or approval can be mistaken for readiness.
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The success condition

Every important state must show its evidence, uncertainty and accountable decision-maker. Unknown must remain unknown until it is resolved.

PROPOSED ANSWER

The solution and its control principles

Principle	Meaning in this capstone
Evidence before status	A label is not truth. A state must cite current, permitted evidence.
Explicit unknowns	Missing or conflicting evidence becomes UNKNOWN, not automatic approval.
Human authority	Identity and Quality release require an authorized human decision.
Safe side effects	Commands are idempotent; uncertain outcomes are reconciled before retry.
Owned exceptions	Conflicts become cases with an owner, severity and resolution trail.
Auditability	Important actions and decisions enter a tamper-evident audit chain.
AI independence	The deterministic core works with AI off; AI can only recommend.

The Control Tower does not replace source systems or decision authorities. It coordinates evidence, state, exceptions and controlled actions across them.

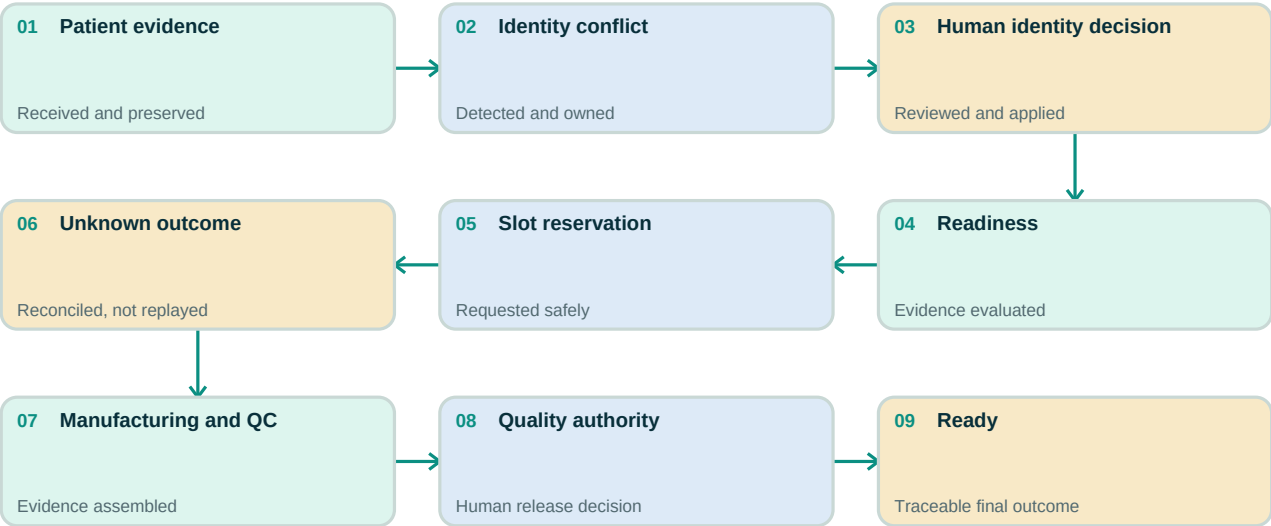
What users see

Journey Enrollment through identity, slot, manufacturing, Quality and readiness.	Controls Human authorization, idempotency, audit validity and AI mode.	Assurance Stage, test, evaluation and requirement results with restrictions.
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ONE PATH - NO DEVIATION

End-to-end patient-to-batch flow

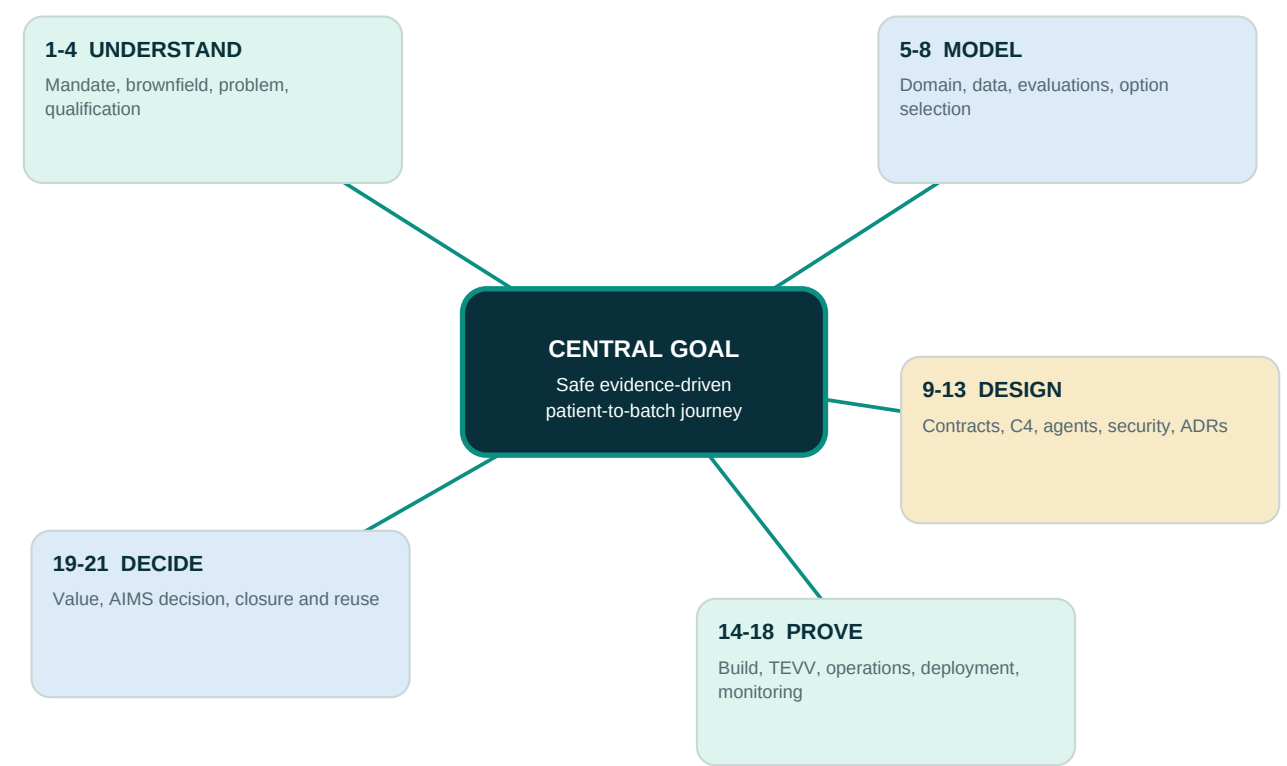
Always explain the capstone in this order. Do not begin with AI or architecture. Begin with the patient journey and the evidence problem.



Decision point	Unsafe shortcut avoided	Controlled response
Identity	Conflicting evidence creates a case	Authorized human resolution
Reservation	Timeout creates an unknown result	Reconcile before any repeat
Release	Evidence exists but authority is absent	Quality decision remains mandatory

NAVIGATION MODEL

The 21-stage FDE mind map



Stages	Question to remember	Examples
1-4	Why are we doing this?	Mandate, brownfield, problem, value, regulation
5-8	What must the solution understand?	DDD, data, evaluations, option selection
9-13	How should we design and govern it?	Contracts, C4, agents, security, ADRs
14-18	Did we build, test and operate it?	Code, POCs, TEVV, runbooks, deployment, monitoring
19-21	What was proven and what happens next?	Value, AIMS, CAPA, lifecycle, retirement, reuse

Memory sentence: Understand -> Model -> Design -> Prove -> Decide

WHY ARE WE DOING THIS?

Stages 1-4: Understand and frame

Stage	Activity	What we did	Key evidence
1	Mandate and field immersion	Define mission, scope, stakeholders, affected groups and decision rights.	Charter, evidence policy, stakeholder map, RACI
2	Discover process and architecture	Parse the supplied ZIP, reproduce the baseline and map current process, systems and risks.	132-file inventory, forensic baseline, current-state architecture
3	Frame problem, root cause and value	Use SCQA, root-cause analysis, KPIs and CTQs to prioritize the real problem.	Problem frame, root causes, KPI baseline, success criteria
4	Triage regulation and qualify use case	Define intended purpose, AI suitability, prohibited uses and human authority.	Use-case cards, boundaries, G1 decision

Typical questions

- What was the client asking for? -> Stages 1-3.
- How did you understand the legacy system? -> Stage 2.
- How did you define the problem and KPIs? -> Stage 3.
- Did you consider regulation and responsible AI before building? -> Stage 4.

Stage 4 rule: no application architecture is selected until domain, data, evaluation and option work in Stages 5-8 is complete.

WHAT MUST THE SOLUTION UNDERSTAND?

Stages 5-8: Model and select

Stage	Activity	What we did	Key evidence
5	Model the domain	Create the ubiquitous language, bounded contexts, aggregates, ownership, rules, events and state machines.	DDD model and machine-readable domain specification
6	Qualify data and knowledge	Assess quality, lineage, provenance, permissible use, representativeness and knowledge gaps.	Profiles, inventories, source-trust and gap registers
7	Define evaluations, impacts and risks	Freeze normal, failure, security, recovery and human-oversight cases before implementation.	57-case catalog, thresholds, risk and impact assessment
8	Generate, test and select options	Compare six options using spikes, TCO, provider/build-buy and weighted trade-offs.	Selected deterministic evidence-driven Control Tower

DDD answer

Yes. DDD was applied in Stage 5 through the glossary, bounded contexts, aggregates, business rules, domain events and canonical state machines.

Why this matters

DDD separates domain authority from software convenience. Clinical, manufacturing, scheduling and Quality meanings are explicit, so one technical service cannot silently redefine another function's truth.

HOW SHOULD WE BUILD AND GOVERN IT?

Stages 9-13: Design and approve

Stage	Activity	What we did	Key evidence
9	Information architecture	Define evidence, event, command and recommendation contracts.	Three versioned JSON schemas
10	Application and AI architecture	Create C4 context, container and component views plus API and failure design.	Target C4 and bounded AI architecture
11	Agentic orchestration	Assess agent suitability, decision authority, termination, escalation and fallbacks.	Bounded workflow; no autonomous consequential decisions
12	Security, guardrails and suppliers	Threat-model access, privacy, injection, evidence integrity and supplier dependency.	Threat model, controls, SBOM, AIBOM and exit plan
13	ADRs and delivery specification	Approve requirements, NFRs, ADRs, backlog, traceability and gate conditions.	27 requirements, 12 ADRs and G3 decision

C4 answer

Yes. Stage 2 records the current state, Stage 10 defines target context/container/component views, and Stage 14 records the as-built architecture.

Spec-driven development answer

Yes. Contracts, requirements, ADRs, evaluation cases, controls and acceptance criteria were defined before the Stage 14 implementation.

DID IT WORK UNDER CONTROL?

Stages 14-18: Build, assure and operate

Stage	Activity	What we did	Key evidence
14	Engineer and integrate	Build the FastAPI service, browser Control Tower, evidence store, services, adapters, audit and three POCs.	Working application and as-built architecture
15	Evaluate, attack and assure	Run automated, integration, E2E, adversarial, performance and residual-risk evaluation.	77 tests; 55 pass; 0 fail; 2 inconclusive human studies
16	Prepare operations and recovery	Define SLOs, telemetry, alerts, runbooks, incidents, backup, restore, training and change control.	Operational evidence and G5 decision
17	Deploy progressively	Simulate shadow, canary, rollback and adoption evidence.	20/20 shadow matches; 10/10 canary success
18	Monitor and validate resilience	Simulate dashboards, drift, incidents, alerts and controlled-chaos recovery.	Monitoring and resilience report

Important: deployment and operation were simulated locally. This is evidence of POC control feasibility, not production history.

Stage 15 evidence location

docs/stages/stage_15 contains the TEVV report, red-team report, residual-risk decisions, G4 record, stage review, evaluation results, performance results and test summary.

WHAT WAS PROVEN AND WHAT HAPPENS NEXT?

Stages 19-21: Prove, decide and close

Stage	Activity	What we did	Key evidence
19	Prove value and tell the story	Compare technical evidence with the baseline and separate measured results from hypotheses.	Value report and executive decision story
20	Evaluate AIMS and lifecycle	Audit lifecycle controls, identify gaps, create CAPAs and make the G6 decision.	Seven open CAPAs; RESTRICT AND CHANGE
21	Retire and capture reusable IP	Define retirement controls, close the academic project and retain reusable patterns.	Closure record, lessons and reusable-IP catalog

What was proven

Control feasibility Evidence-backed states, explicit unknowns and human authority work on synthetic journeys.	Engineering quality Tests, evaluations, rollback, audit and traceability are reproducible.	Decision discipline Known limitations become CAPAs and restrictions, not inflated claims.
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What was not proven

Clinical benefit, production safety, regulatory compliance, human-factor performance, real-world savings, live integrations and live-model AI quality were not established.

Final lifecycle decision: RESTRICT AND CHANGE

WHAT THE LIVE DEMO PROVES

The three proofs of concept

POC	Business area	Problem	Controlled response	Result
POC 1	Identity and readiness	Conflicting MRN and DOB evidence	Open an owned case; authorized human applies reviewed proposal	Identity APPLIED; readiness SATISFIED; 5 citations
POC 2	Safe slot orchestration	External reservation succeeds but response times out	Record OUTCOME_UNKNOWN; reconcile; block second dispatch	Reconciled to SUCCEEDED; duplicate prevented
POC 3	Quality release	Manufacturing, QC, deviation and thermal evidence exists without authority	Keep release UNKNOWN until authorized Quality decision	HUMAN AUTHORIZED; final outcome SATISFIED

The common pattern

Detect uncertainty -> Preserve evidence -> Assign authority -> Apply a controlled decision -> Recalculate state -> Record audit

What the POCs deliberately avoid

- No identity is invented from a similarity score.
- No timeout is treated automatically as failure.
- No duplicate external request is sent without reconciliation.
- No missing Quality decision is treated as release.
- No AI recommendation changes authoritative state.

DDD, C4, CONTRACTS AND ADRS

Architecture quick guide

Method	Stage	Purpose	Project evidence
DDD	Stage 5	Defines what the business domain means	Glossary, bounded contexts, aggregates, rules, events and state machines
C4	Stages 2, 10, 14	Shows how the technical system is structured	Current state, target context/container/component and as-built views
Contracts	Stage 9	Defines stable information exchanged between parts	Command, event-envelope and recommendation schemas
Agent design	Stage 11	Defines autonomy, tools, sequencing and escalation	Bounded deterministic orchestration with human handoff
Threat model	Stage 12	Identifies how controls could fail or be abused	Security, privacy, prompt-injection and supplier controls
ADRs	Stage 13	Records why major design choices were made	Twelve decisions with alternatives and consequences

One-sentence relationship

DDD defines the domain. C4 structures the system. Contracts define exchanges. ADRs explain choices. Tests prove behavior.

Core implementation layers

Interface Browser Control Tower, FastAPI endpoints and CLI.	Domain services Evidence, readiness, commands, cases, decisions and audit.	Infrastructure SQLite persistence and simulated external adapters.
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THE SAFEST ANSWER

AI and human authority

The application is useful with AI off. AI is optional, non-authoritative and replaceable.

Mode	Status	Meaning
off	Recommended default	No model is called. Deterministic domain controls execute normally.
fake	Bounded demonstration	A local deterministic adapter returns a non-binding recommendation.
live	Not supported	No external model or provider is connected or evaluated.

AI cannot

- establish patient identity or chain of identity;
- make clinical eligibility or readiness decisions;
- execute an external reservation independently;
- approve a deviation or product disposition; or
- release the product.

Authorized humans retain

- identity-resolution authority;
- clinical and operational exception decisions;
- Quality disposition and release authority; and
- the right to override, abstain and escalate with recorded rationale.

Reviewer answer: We did not add AI to create authority. We bounded AI so it can assist without becoming authority.

HOW CLAIMS ARE CONTROLLED

Evidence, traceability and assurance

Evidence -> Root cause -> Requirement -> ADR -> Code -> Test or evaluation -> KPI -> Lifecycle decision

Traceability element	Stage	Evidence
Source evidence	Stage 2	Frozen ZIP digest and 132 extracted-file hashes
Problem and CTQs	Stage 3	Root causes, baseline and success/failure criteria
Business rules	Stage 5	Eleven safety and authority rules
Evaluation design	Stage 7	Fifty-seven frozen cases and thresholds
Design decision	Stages 9-13	Contracts, C4, controls, requirements and ADRs
Implementation	Stage 14	Source code and three integrated POCs
Verification	Stage 15	Tests, TEVV, adversarial and performance evidence
Lifecycle conclusion	Stages 19-21	Value boundary, CAPAs, decision and closure

Why this matters

A reviewer can start from a business claim and follow it to the source evidence, design decision, implementation and test. VERIFIED_INTERNAL_POC never means production validated.

RESULTS WITHOUT OVERCLAIMING

What the evidence says

Measure	Result	Correct interpretation
FDE stages	21/21	Every stage contains artifacts
Automated tests	77/77 passed	Unit, integration, E2E, recovery, security and performance
Evaluation catalog	57 executed	55 pass; 0 fail; 2 inconclusive human studies
Requirements	25/27 internal	Two human-factor requirements need external evidence
Shadow simulation	20/20 matched	AI-off and fake modes preserved domain outcomes
Canary simulation	10/10 succeeded	Synthetic local journeys only
Recovery	Passed	Backup, restore and AI-off rollback simulation
Audit	Valid	Tamper-evident chain verified in the demo

Two inconclusive cases

Automation bias and safe human override require controlled participants. Software tests cannot prove how real people will respond, so the evidence correctly remains inconclusive.

Best answer: We have strong internal technical evidence and clearly identified external evidence gaps. We do not convert gaps into passes.

WHAT TO SAY AND SHOW

Five-minute live demo runbook

Time	Show	Say
0:00	Opening screen	This is one controlled journey from a patient to that patient's batch.
0:30	Four controls	Identity is human-authorized; orchestration is idempotent; Quality retains authority; AI is off.
1:00	Run demo	Select AI off - recommended, then click Run demo.
1:20	POC 1	A conflict becomes an owned case. A human resolves identity; readiness becomes satisfied with citations.
2:10	POC 2	A timeout becomes unknown, is reconciled to success and is never dispatched twice.
3:00	POC 3	Evidence informs release, but only Quality can authorize it.
3:45	Evidence console	Show ten references, valid audit, deterministic mode, metrics and state digest.
4:20	Assurance	Show 21 stages, 77 tests, 55 of 57 evaluations and 25 of 27 requirements.
4:45	Decision	Close with Restrict and Change and the seven CAPAs before any real pilot.

Starting locally

Double-click START_DEMO.command, keep the Terminal open and use `http://127.0.0.1:8000`. The launcher creates `.venv` when needed and starts with `AI_MODE=off`.

Do not begin the demo with code. Begin with the patient, the evidence problem and the accountable decision-maker.

QUESTION ROUTER - PART 1

Reviewer Q&A;: discovery, domain and data

Question	Stage	Short answer
What was the ask?	1-3	One trustworthy, evidence-backed view of identity, readiness, reservation and release.
How did you inspect the legacy estate?	2	Read-only forensic baseline, process/system mapping, defects, workarounds and dependencies.
Did you preserve source evidence?	2	Yes. The ZIP digest and all 132 frozen extraction hashes are verified.
How did you frame the problem?	3	SCQA, root causes, waste, KPIs, CTQs and explicit success/failure criteria.
Did you use DDD?	5	Yes: ubiquitous language, bounded contexts, aggregates, ownership, rules, events and state machines.
How do you handle missing data?	5	Missing or conflicting evidence returns UNKNOWN; it never creates automatic approval.
How do you handle late data?	5-6	Bitemporal evidence records both occurrence time and system-recorded time.
How did you assess data?	6	Quality, lineage, provenance, access, representativeness, knowledge and gap profiling.
How did you choose a source of truth?	6	Trust is attribute-specific; source assertions are preserved rather than globally overwritten.

QUESTION ROUTER - PART 2

Reviewer Q&A;: evaluation and architecture

Question	Stage	Short answer
Were evaluations defined before code?	7	Yes. Fifty-seven cases and thresholds were frozen before Stage 14 implementation.
Did you compare options?	8	Yes. Six options were compared using spikes, TCO, provider/build-buy and weighted scoring.
Did you use contracts?	9	Yes. Commands, events and recommendations use versioned JSON schemas.
Did you use C4?	2, 10, 14	Yes. Current-state, target and as-built architecture views are retained.
Did you design for failure?	10	Yes. Timeouts, partial effects, unknown outcomes, reconciliation, compensation and fallback are explicit.
Did you use agents?	11	Agentic suitability was assessed, but consequential autonomy was deliberately rejected.
Did you perform threat modelling?	12	Yes. Access, privacy, prompt injection, evidence integrity and supplier risks were treated.
Do you have SBOM and AIBOM?	12, 14	Yes, as academic evidence; they are not signed production attestations.
Did you use ADRs?	13	Yes. Twelve ADRs record major choices, alternatives, consequences and reversibility.

QUESTION ROUTER - PART 3

Reviewer Q&A;: build, test and operations

Question	Stage	Short answer
What did you build?	14	A FastAPI service, browser Control Tower, SQLite evidence store, services, adapters, audit and three POCs.
How do you prevent duplicate booking?	5, 10, 14	Payload-bound idempotency plus reconciliation before retry.
Did you test the solution?	15	Yes. All 77 tests pass; 55 evaluation cases pass; zero fail; two need human studies.
Did you red-team it?	15	Yes. Adversarial, injection, authority, conflict, outage, recovery and unknown-outcome cases are included.
Why are two cases inconclusive?	15	Automation bias and override usability require controlled human participants.
Do you have runbooks?	16	Yes. SLOs, alerts, incidents, change, backup, restore, rollback and training are documented.
Was it deployed?	17	Only as simulation: 20 shadow matches and 10 successful synthetic canary journeys.
How is it monitored?	18	Dashboard, SLO, alert, drift, incident and resilience-drill specifications are retained.

QUESTION ROUTER - PART 4

Reviewer Q&A;: AI, value and lifecycle

Question	Stage	Short answer
Did you use AI?	10-14	AI is optional and off by default. A local fake adapter demonstrates only a bounded recommendation.
Can AI release the product?	4, 11, 12	No. Quality release always requires authorized human authority.
What value was proven?	19	Technical control feasibility: visibility, duplicate prevention, ownership and traceability.
Did you prove ROI?	19	No. Real savings, outcomes and production TCO were not measured.
Did you use AIMS?	20	An AIMS lifecycle review was performed, but no certification is claimed.
Is it production-ready?	20	No. The formal decision is RESTRICT AND CHANGE.
What remains?	20	Seven CAPAs, human studies, enterprise controls, live integrations and independent assurance.
How was the project closed?	21	Retirement, lessons learned, reusable IP and final closure were documented.

When challenged, do not defend an unsupported claim. Point to the stage, evidence and limitation.

REPOSITORY MAP

Where to find the evidence

Need	Location
Client ask and demo story	docs/CLIENT_ASK_AND_END_TO_END_DEMO.md
Five-minute presenter guide	docs/DEMO_GUIDE.md
Final capstone summary	docs/FINAL_CAPSTONE_REPORT.md
All 21 stage folders	docs/stages/stage_01 through docs/stages/stage_21
DDD artifacts	docs/stages/stage_05
Data and knowledge artifacts	docs/stages/stage_06
Evaluation design	docs/stages/stage_07
C4 target architecture	docs/stages/stage_10/01_APPLICATION_AND_AI_ARCHITECTURE.md
Security and guardrails	docs/stages/stage_12
ADRs and requirements	docs/stages/stage_13 and requirements/
As-built implementation evidence	docs/stages/stage_14 and src/fde_capstone/
TEVV and red-team evidence	docs/stages/stage_15
Operations and recovery	docs/stages/stage_16
Shadow/canary simulation	docs/stages/stage_17
Monitoring and resilience	docs/stages/stage_18
Value and executive story	docs/stages/stage_19
AIMS, CAPAs and decision	docs/stages/stage_20
Closure and reusable IP	docs/stages/stage_21
Tests	tests/ and reports/generated/pytest-results.xml
Final machine verification	evidence/final_verification.json

Repository

github.com/86sunbot/FDE-Final-Capstone

WHAT NOT TO CLAIM

The honest boundary

Demonstrated Synthetic technical control feasibility, traceability, testing, audit and recovery.	Restricted Real patient data, consequential AI, automated release and production integrations.	Still required Human studies, enterprise controls, live integration, validation and independent assurance.
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Do not say

- The system is clinically validated or regulator approved.
- The application is production-ready.
- The project proved patient benefit, savings or ROI.
- The demo principals are enterprise IAM or electronic signatures.
- The external systems and decision authorities are live.
- A live AI model was evaluated.

Say instead

The capstone demonstrates a reproducible synthetic POC with strong internal technical evidence. It identifies the external evidence and controls required before any real pilot.

RESTRICT AND CHANGE: accept the academic POC, keep AI off, prohibit real-data use and close seven CAPAs before a new accountable gate sequence.

SAY IT SIMPLY

Ready-to-use talk tracks

Thirty-second answer

This capstone creates one evidence-driven patient-to-batch view for Cell and Gene Therapy. It resolves identity conflicts through human authority, prevents duplicate manufacturing reservations after uncertain responses, and keeps product release with Quality. All core controls work with AI off. The synthetic POC passed 77 tests, but production use remains restricted.

One-minute answer

CGT treatments in this capstone are manufactured for a specific patient, so identity, custody, timing and release must remain traceable. The current environment spreads evidence across clinical, logistics, manufacturing and Quality systems. We followed all 21 FDE stages to understand the brownfield system, model the domain and data, predefine evaluations, select a controlled architecture, build three POCs and test the result. The demo shows human-authorized identity resolution, safe reconciliation of an unknown reservation outcome and Quality-authorized release. AI is optional and off by default. The POC is technically successful, but real-data and production use remain prohibited until external evidence and seven CAPAs are completed.

Closing answer

Evidence before status. Human authority before automation. Unknown remains unknown until evidence and the correct authority resolve it.

TERMS REVIEWERS MAY USE

Essential glossary

Term	Simple meaning
ADR	Architecture Decision Record: why a major design choice was made.
AIBOM	AI Bill of Materials: models, adapters, modes and AI dependencies.
AIMS	AI Management System: lifecycle governance, controls, audit and improvement.
Bitemporal	Records both when something happened and when the system learned it.
Bounded context	A domain boundary within which terms, rules and authority are consistent.
C4	Architecture views covering context, containers and components.
CAPA	Corrective and Preventive Action required to close a control gap.
Chain of custody	Traceable possession and handling of patient material or product.
Chain of identity	Traceable link between the patient, collected material and final product.
CTQ	Critical to Quality: a measurable condition essential to a safe result.
DDD	Domain-Driven Design: modelling software around domain language, rules and ownership.
Idempotency	Repeating the same command cannot create a second unintended effect.
SCQA	Situation, Complication, Question, Answer problem-framing structure.
SBOM	Software Bill of Materials: application software dependencies.
TEVV	Testing, Evaluation, Verification and Validation.
UNKNOWN	A deliberate state used when evidence cannot support satisfied or not-satisfied.

YOUR FINAL CHECK BEFORE ANY DISCUSSION

One-page memory sheet

Topic	Answer
Client ask	Trustworthy identity, readiness, reservation, release and evidence
Problem	Fragmented evidence, unsafe assumptions, duplicate effects and unclear authority
Solution	Evidence-driven Control Tower with explicit unknowns and owned exceptions
POC 1	Human-authorized identity resolution and readiness
POC 2	Unknown-outcome reconciliation and duplicate prevention
POC 3	Quality-authorized product release
DDD	Stage 5
C4	Stages 2, 10 and 14
Evaluations	Stage 7 defines; Stage 15 executes
Security	Stage 12 designs; Stage 15 tests
Implementation	Stage 14
Operations	Stages 16-18
Value	Stage 19 - technical feasibility, not ROI
Decision	Stage 20 - RESTRICT AND CHANGE
Closure	Stage 21
Evidence	21 stages; 77 tests; 55/57 pass; 25/27 internal requirements
AI	Optional, off by default and never authoritative
Boundary	Synthetic academic POC; not clinical, regulatory or production validation

UNDERSTAND -> MODEL -> DESIGN -> PROVE -> DECIDE

Surya's final sentence

We built a controlled, evidence-driven CGT patient-to-batch POC that preserves human authority, manages uncertainty and proves its internal technical behavior without claiming production readiness.